



BobSim

TransientEval Lateral Transient Response

Step response and sustained-sine FRF

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Run Notes

- Representative step steer response
- Continuous sinusoidal responses
- Rise time uses 50 percent input to 90 percent response thresholds
- Frequency delays are reported at 0.5 Hz and 1.0 Hz
- Gain and phase slopes summarize the FRF trend
- Steering shown in degrees for readability

TransientEval Metrics Summary 15.0 m/s

Time Domain - Step Response

STEP RESPONSE

a_y Rise Time (50-90%)	0.05	s
Sideslip Rise Time (50-90%)	0.17	s
Yaw Rise Time (50-90%)	0.06	s
Yaw Peak Response Time	0.10	s
a_y DC Gain	29.92	$\frac{\text{m/s}^2}{\text{rad}}$
Sideslip DC Gain	0.040	rad/rad
Yaw DC Gain	1.97	(rad/s)/rad
Roll DC Gain	0.076	rad/rad

STEADY-STATE / OVERSHOOT

a_y Peak	3.03	m/s^2
a_y Steady-State	2.52	m/s^2
a_y Overshoot	20.5	%
Yaw Steady-State	0.169	rad/s
Yaw Overshoot	0.025	rad/s
Roll Steady-State	0.007	rad
Roll Overshoot	0.001	rad
Settling Time	1.21	s

TransientEval Metrics Summary 15.0 m/s

Frequency Domain - Sustained Sine

FREQUENCY RESPONSE - CORE

DC Gain (a_y)	29.92	$\frac{\text{m/s}^2}{\text{rad}}$
DC Gain (r)	1.97	(rad/s)/rad
Peak Gain (a_y)	31.07	$\frac{\text{m/s}^2}{\text{rad}}$
Peak Freq (a_y)	1.00	Hz
Bandwidth (-3 dB)	1.00	Hz
Gain Slope (a_y)	1.07	dB/dec
Gain Slope (r)	0.19	dB/dec
Phase Slope (a_y)	7.34	deg/dec
Phase Slope (r)	16.27	deg/dec

DELAY / COUPLING

Ay/Yaw Delay @ 0.5 Hz	0.008	s
Yaw/Steer Delay @ 0.5 Hz	0.027	s
Difference @ 0.5 Hz	-0.019	s
Ay/Yaw Delay @ 1.0 Hz	0.010	s
Yaw/Steer Delay @ 1.0 Hz	0.027	s
Difference @ 1.0 Hz	-0.017	s
Gain Variation	3.8	%
Fit Error (a_y)	7.64e-03	-
Fit Error (r)	5.23e-03	-

TransientEval Metrics Summary 20.0 m/s

Time Domain - Step Response

STEP RESPONSE

a_y Rise Time (50-90%)	0.09	s
Sideslip Rise Time (50-90%)	0.08	s
Yaw Rise Time (50-90%)	0.06	s
Yaw Peak Response Time	0.10	s
a_y DC Gain	46.54	$\frac{\text{m/s}^2}{\text{rad}}$
Sideslip DC Gain	-0.002	rad/rad
Yaw DC Gain	2.32	(rad/s)/rad
Roll DC Gain	0.121	rad/rad

STEADY-STATE / OVERSHOOT

a_y Peak	4.25	m/s^2
a_y Steady-State	3.90	m/s^2
a_y Overshoot	9.0	%
Yaw Steady-State	0.197	rad/s
Yaw Overshoot	0.021	rad/s
Roll Steady-State	0.011	rad
Roll Overshoot	0.001	rad
Settling Time	1.22	s

TransientEval Metrics Summary 20.0 m/s

Frequency Domain - Sustained Sine

FREQUENCY RESPONSE - CORE

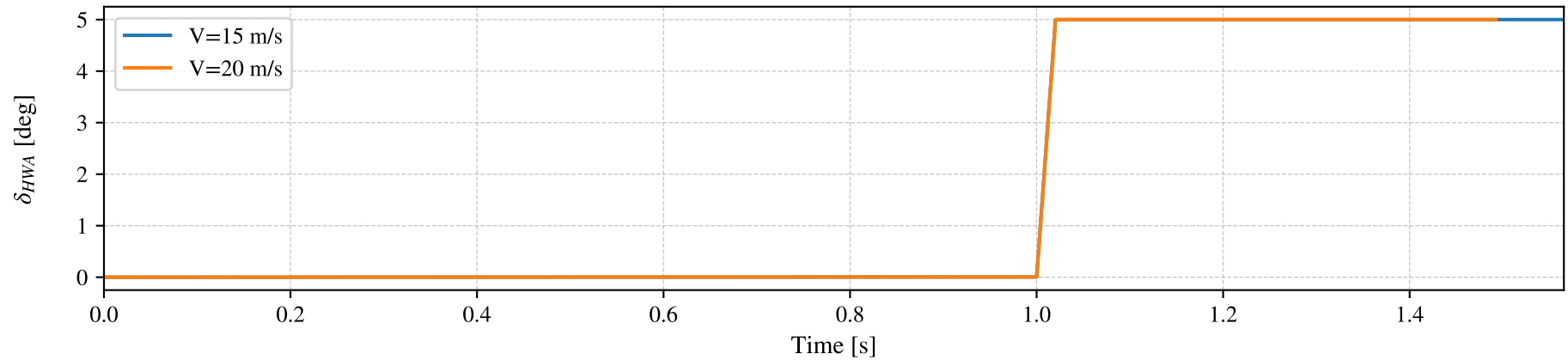
DC Gain (a_y)	46.54	$\frac{\text{m/s}^2}{\text{rad}}$
DC Gain (r)	2.32	(rad/s)/rad
Peak Gain (a_y)	46.78	$\frac{\text{m/s}^2}{\text{rad}}$
Peak Freq (a_y)	1.00	Hz
Bandwidth (-3 dB)	1.00	Hz
Gain Slope (a_y)	0.15	dB/dec
Gain Slope (r)	0.02	dB/dec
Phase Slope (a_y)	20.18	deg/dec
Phase Slope (r)	17.54	deg/dec

DELAY / COUPLING

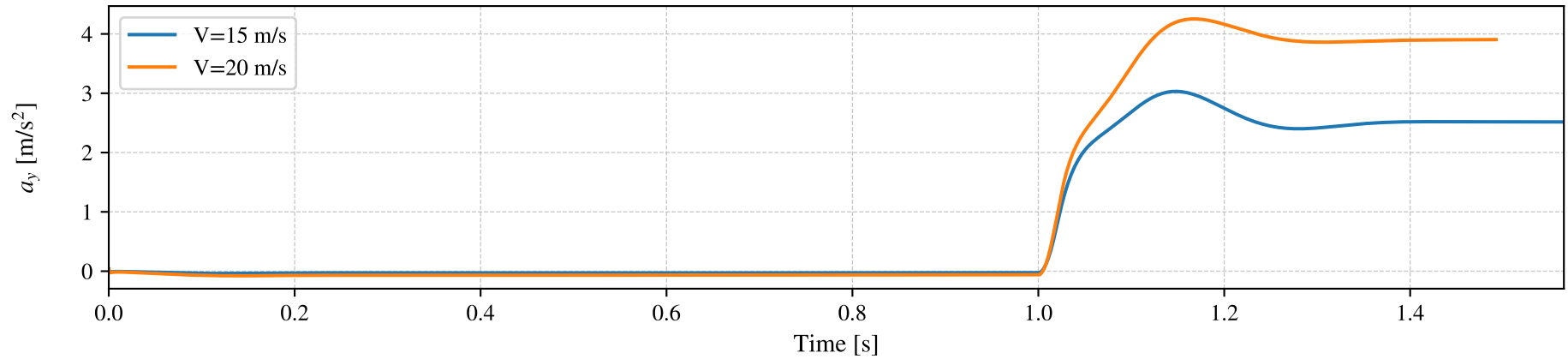
Ay/Yaw Delay @ 0.5 Hz	0.032	s
Yaw/Steer Delay @ 0.5 Hz	0.029	s
Difference @ 0.5 Hz	0.002	s
Ay/Yaw Delay @ 1.0 Hz	0.033	s
Yaw/Steer Delay @ 1.0 Hz	0.030	s
Difference @ 1.0 Hz	0.003	s
Gain Variation	0.5	%
Fit Error (a_y)	7.76e-03	-
Fit Error (r)	3.39e-03	-

Step Steer Response (TransientEval)

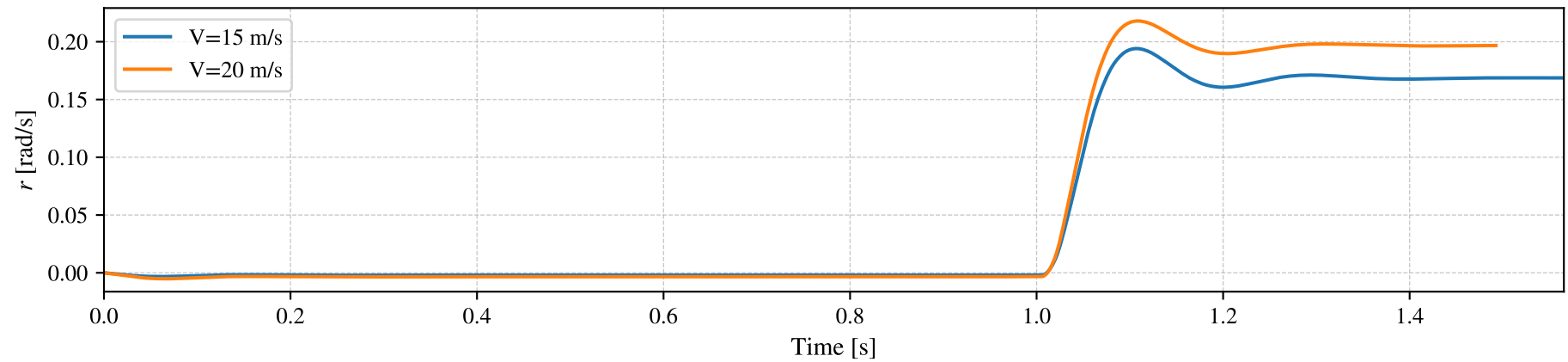
Steering Wheel Angle δ_H



Lateral Acceleration a_y

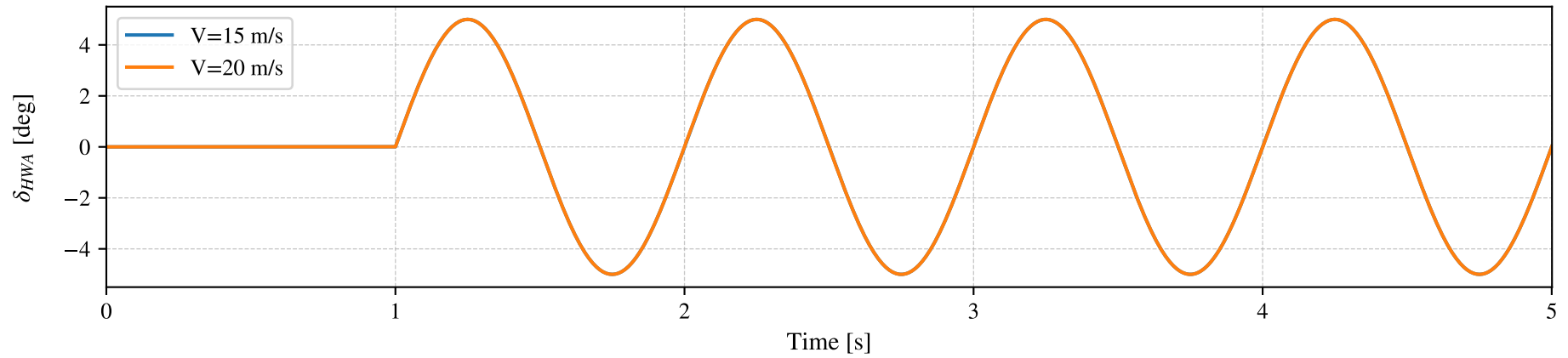


Yaw Velocity r

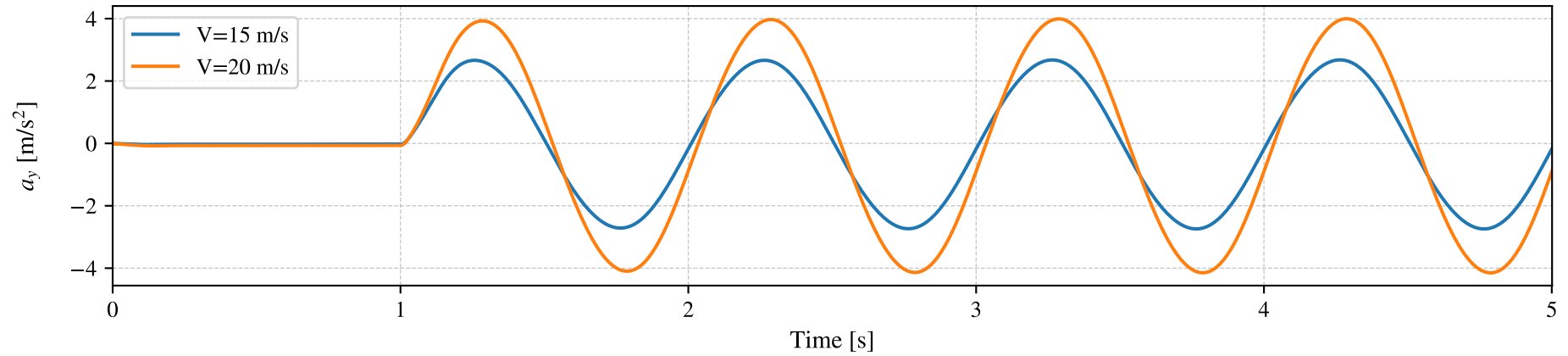


Continuous Sinusoidal Response

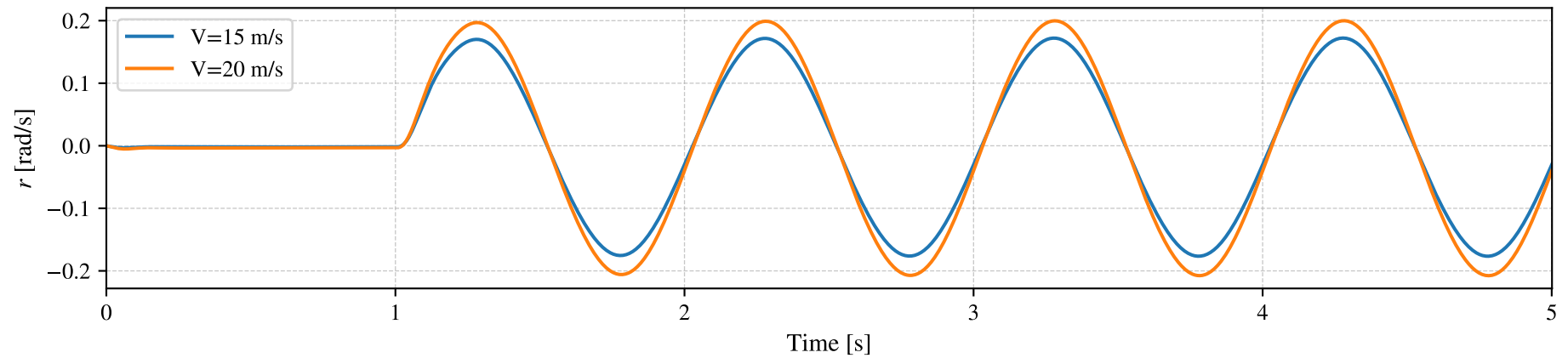
Steering Wheel Angle δ_H



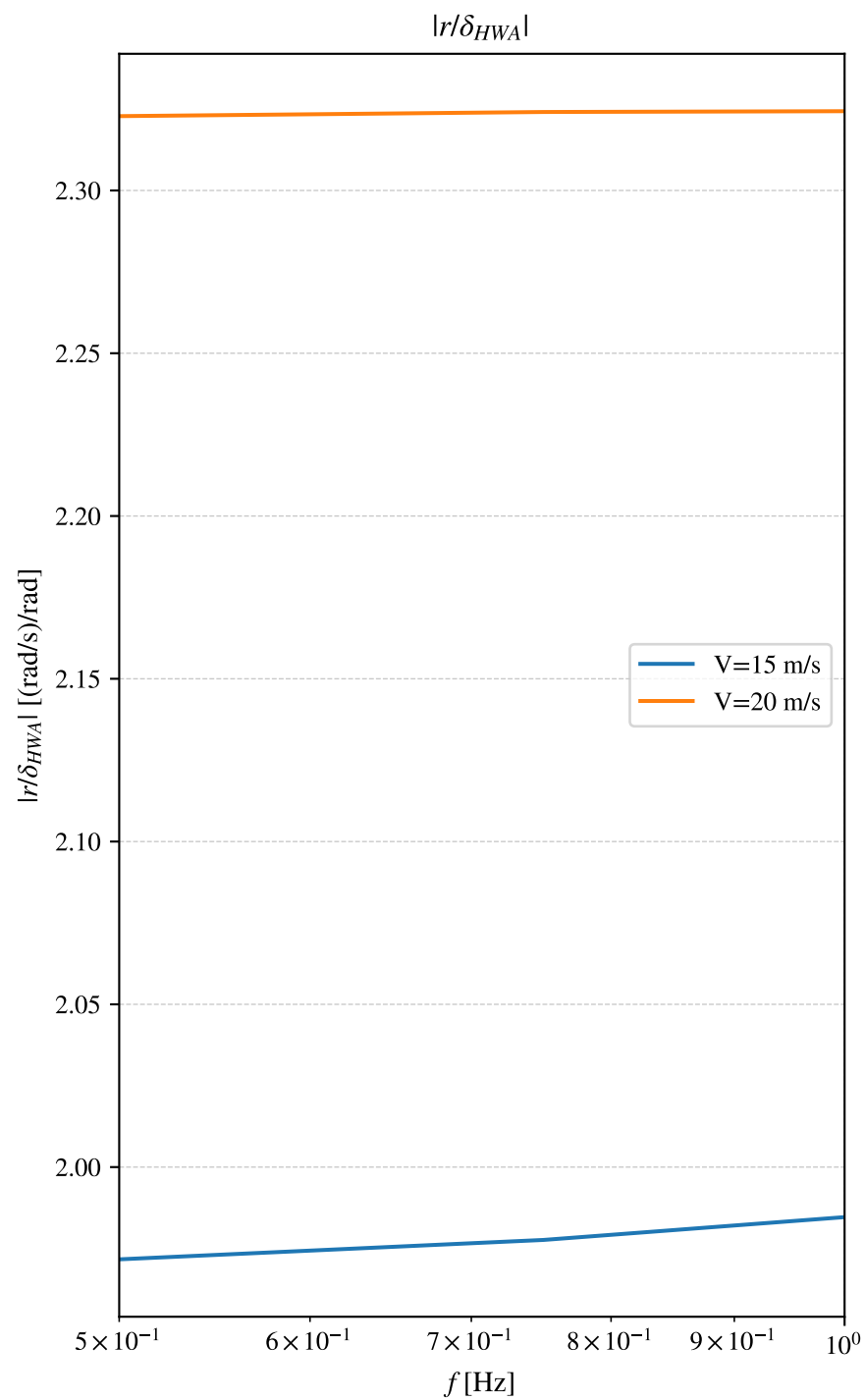
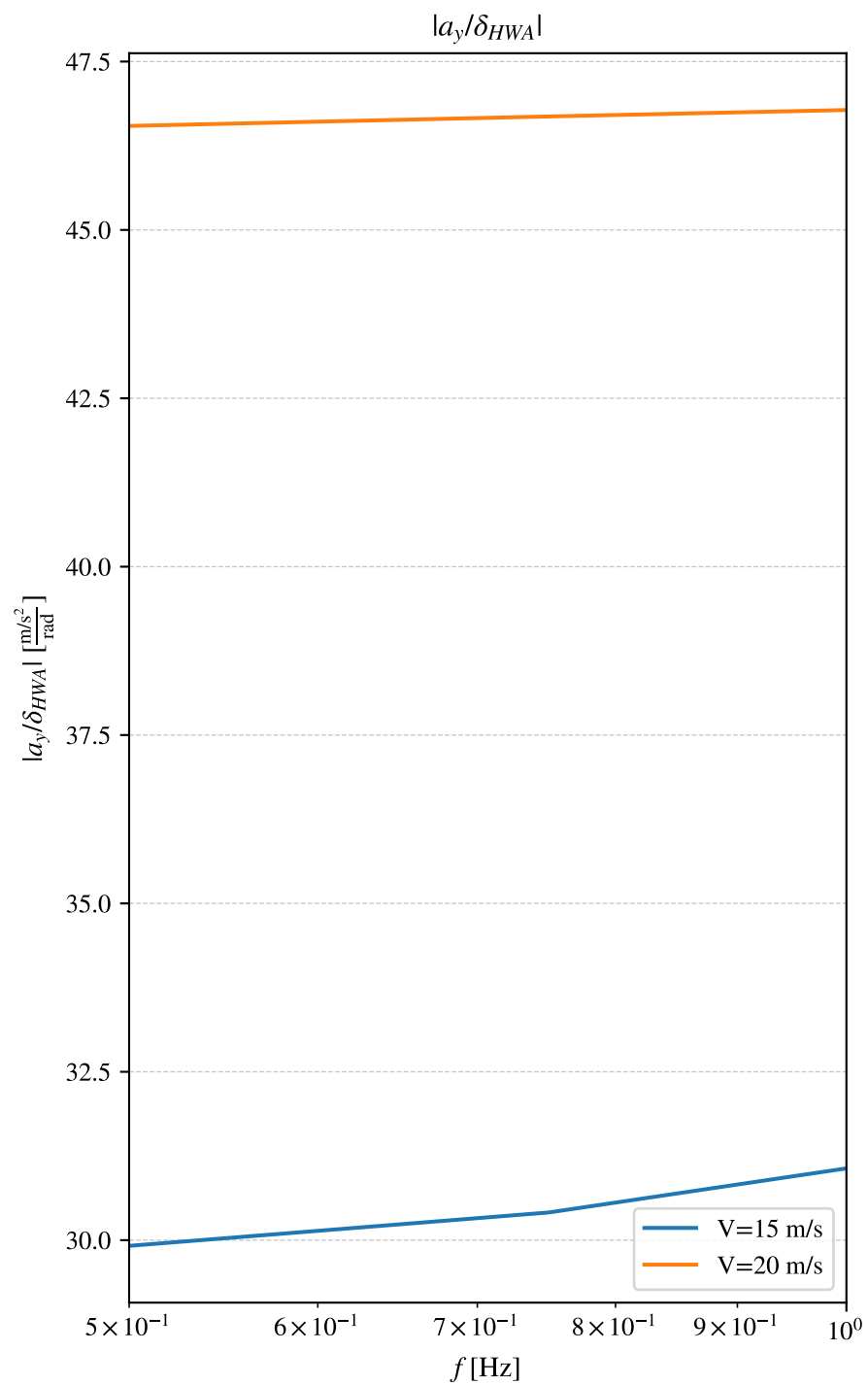
Lateral Acceleration a_y



Yaw Velocity r

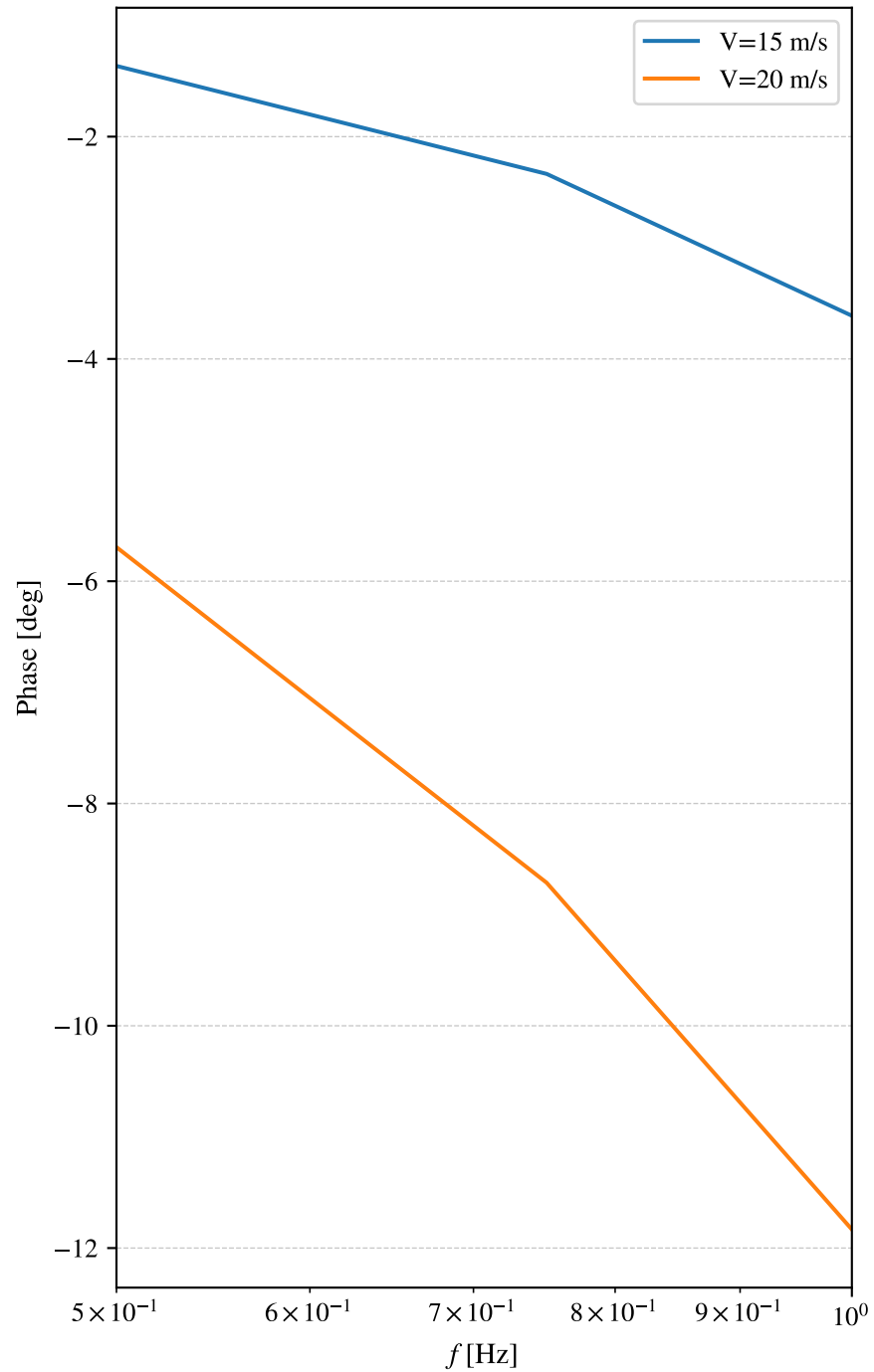


Frequency Response — Gain (TransientEval)



Frequency Response — Phase (TransientEval)

Phase: a_y vs δ_{HWA}



Phase: r vs δ_{HWA}

